

Highway SVM Audit Summary

Prepared by Cyfrin

Version 2.1

Lead Auditors

Farouk

qpzm

September 29, 2026

Contents

1 About Cyfrin **2**

2 About This Document **2**

3 Disclaimer **2**

4 Copyright and License **2**

5 Audit Overview **3**

6 Audit Scope **3**

7 Risk Classification **5**

8 Protocol Summary **5**

8.1 Components 5

8.2 Message Lifecycle 6

9 Attestation **6**

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Highway on Highway SVM. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.1 dated September 29, 2026, was delivered to Highway, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Highway or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 10 days, the Cyfrin team conducted an audit on the Highway SVM code provided by Highway. The findings consist of 1 Critical, 1 High, 3 Medium & 40 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Highway SVM
Repository	hway-solana
Commit	b8246bbd6d68...
Audit Timeline	July 20th - July 31st, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	1	1	0
Medium Risk	3	3	0
Low Risk	40	34	6
Informational	33	28	5
Gas Optimizations	2	2	0
Total	80	69	11

6 Audit Scope

The audit scope was limited to:

```
programs/config/src/instructions/add_fee_token.rs
programs/config/src/instructions/add_to_whitelist.rs
programs/config/src/instructions/close_destination_fee_token.rs
programs/config/src/instructions/configure_destination_fee_token.rs
programs/config/src/instructions/configure_destination_fee.rs
programs/config/src/instructions/configure_native_inbound.rs
programs/config/src/instructions/configure_native_outbound.rs
programs/config/src/instructions/configure_native_token_bridge.rs
programs/config/src/instructions/configure_token_inbound.rs
programs/config/src/instructions/configure_token_outbound.rs
programs/config/src/instructions/create_token_bridge.rs
programs/config/src/instructions/initialize.rs
programs/config/src/instructions/mod.rs
programs/config/src/instructions/pause.rs
programs/config/src/instructions/register_chain.rs
programs/config/src/instructions/register_program.rs
programs/config/src/instructions/register_token.rs
programs/config/src/instructions/remove_chain.rs
programs/config/src/instructions/remove_fee_token.rs
programs/config/src/instructions/remove_from_whitelist.rs
programs/config/src/instructions/remove_program.rs
```

programs/config/src/instructions/remove_token.rs
programs/config/src/instructions/set_fee_config.rs
programs/config/src/instructions/set_native_token_id.rs
programs/config/src/instructions/set_token_enabled.rs
programs/config/src/instructions/unconfigure_native_token_bridge.rs
programs/config/src/instructions/unconfigure_token_bridge.rs
programs/config/src/instructions/unpause.rs
programs/config/src/instructions/update_admin.rs
programs/config/src/instructions/update_bridge_limits.rs
programs/config/src/instructions/update_bridge_mode.rs
programs/config/src/instructions/update_chain_status.rs
programs/config/src/instructions/update_fee_token.rs
programs/config/src/instructions/update_token_bridge.rs
programs/config/src/states/bridge_configuration.rs
programs/config/src/states/chain_info.rs
programs/config/src/states/config.rs
programs/config/src/states/destination_fee_config.rs
programs/config/src/states/destination_fee_token_config.rs
programs/config/src/states/fee_config.rs
programs/config/src/states/fee_token_config.rs
programs/config/src/states/mod.rs
programs/config/src/states/native_bridge_config.rs
programs/config/src/states/token_info.rs
programs/config/src/states/whitelist.rs
programs/common/src/lib.rs
programs/config/src/utils/decimals.rs
programs/config/src/constants.rs
programs/config/src/utils/mod.rs
programs/config/src/errors.rs
programs/config/src/events.rs
programs/config/src/lib.rs
programs/entry/src/instructions/emit_message.rs
programs/entry/src/instructions/mod.rs
programs/entry/src/states/message_nonce.rs
programs/entry/src/states/mod.rs
programs/entry/src/constants.rs
programs/entry/src/utils/fee_quote.rs
programs/entry/src/errors.rs
programs/entry/src/events.rs
programs/entry/src/lib.rs
programs/entry/src/utils.rs
programs/executor/src/instructions/execute_message.rs
programs/executor/src/instructions/initialize.rs
programs/executor/src/instructions/mod.rs
programs/executor/src/instructions/store_message.rs
programs/executor/src/state/executor_authority.rs
programs/executor/src/state/message.rs
programs/executor/src/state/mod.rs
programs/executor/src/state/transfer_execution.rs
programs/executor/src/utils/bitmap.rs
programs/executor/src/utils/bls_verify.rs
programs/executor/src/utils/committee.rs
programs/executor/src/utils/decimals.rs
programs/executor/src/utils/mod.rs
programs/executor/src/constants.rs
programs/executor/src/errors.rs
programs/executor/src/events.rs
programs/executor/src/lib.rs
programs/registry/src/instructions/add_authorized_updater.rs
programs/registry/src/instructions/add_new_active_set.rs
programs/registry/src/instructions/add_operational_key.rs
programs/registry/src/instructions/add_registration_updater.rs
programs/registry/src/instructions/extend_bls_keys.rs

```
programs/registry/src/instructions/init_bls_keys.rs
programs/registry/src/instructions/initialize.rs
programs/registry/src/instructions/mod.rs
programs/registry/src/instructions/register_relayer.rs
programs/registry/src/instructions/remove_authorized_updater.rs
programs/registry/src/instructions/remove_operational_key.rs
programs/registry/src/instructions/remove_registration_updater.rs
programs/registry/src/instructions/remove_relayer.rs
programs/registry/src/instructions/set_min_valid_epoch.rs
programs/registry/src/instructions/update_active_set_update_interval.rs
programs/registry/src/instructions/update_current_active_set.rs
programs/registry/src/instructions/update_registration_window.rs
programs/registry/src/instructions/update_relayer.rs
programs/registry/src/state/bitmap.rs
programs/registry/src/state/bls_keys.rs
programs/registry/src/state/mod.rs
programs/registry/src/state/registry_state.rs
programs/registry/src/state/relayer.rs
programs/registry/src/utils/bn254_pop.rs
programs/registry/src/utils/mod.rs
programs/registry/src/constants.rs
programs/registry/src/errors.rs
programs/registry/src/events.rs
programs/registry/src/lib.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Highway (the Mosaic Bridge) is a cross-chain token-and-message bridge. This summary covers its Solana implementation; the bridge also has EVM (Ethereum) and Substrate implementations that act as these programs' cross-chain counterparties. A message emitted on a source chain is attested by a committee drawn from a shared relayer set and executed on the destination chain, where it mints or releases tokens and/or dispatches a payload call.

8.1 Components

Program	Role
`entry`	Outbound entrypoint: `emit_message` (fee-quote verification, burn/escrow debit, `message_id` assignment, `MessageEmitted`)
`executor`	Inbound entrypoints: `store_payload` (persist message and payload) and `execute_message` (aggregate signature verification, committee selection, mint/release, payload CPI)
`registry`	Relayer registration, relayer keys, operational keys, registered-relayer bitmap, active-set epochs and rotation

Program	Role
config	Chain and token registration, per-corridor bridge configuration and decimals, fee configuration and fee tokens, payload whitelist
common	Shared constants and the generate_message_id implementation

8.2 Message Lifecycle

1. **Outbound (source).** The user calls emit_message with a target chain, an optional token leg (token, target address, amount) and an optional payload leg (target program, encoded call), plus a signed fee quote. The program debits the funds (burn or escrow), allocates a nonce, computes the message_id, and emits MessageEmitted.
2. **Attestation (off-chain committee).** The relayers seated in the active committee for the message's slot co-sign the message and produce a single aggregate signature plus a committee bitmap.
3. **Inbound (destination).** Delivery on Solana is split across two instructions because a single transaction cannot carry both the payload bytes and the verification work. store_payload persists the message fields and payload into a program account; execute_message then re-derives the message_id, verifies the aggregate signature against the committee selected for that slot, records the execution, and performs the token leg (mint or vault release) and the payload leg (a CPI into the target program).

Each (token_id, chain_id) corridor is configured as burn and mint, escrow and release, or closed.

9 Attestation

Cyfrin attests that it performed a security review of the Highway SVM code provided by Highway.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints OK when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-09-29-cyfrin-highway-solana-v2.1.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Highway has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints OK when the copy matches. If the copy was renamed, set F to its file name.

```
H=103c14f1d9f352a3b1160eb1d886120b519e664c0eb6782ba1fd12b288ab3d41
F=2026-09-29-cyfrin-highway-solana-v2.1-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, sha256sum -c can be used in place of shasum -a 256 -c. On Windows, run Get-FileHash <file> -Algorithm SHA256 in PowerShell and compare the result with the hash.